

ITC-101
Examination – June - 2023

B.Tech. II Sem : CE, AE, EE, EI, EC, CSE, IT, AIADS, CSE(BC), IoT
Python Programming

Time : 2 Hrs

Max. Marks : 60
Min. Marks : 19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c & d has internal Choice. Assume missing data, if any.

Word limit be observed as follows:

Part a & b – Objective type/Fill in the blanks/One sentence answer.

Part c – Max 100 words and Part d – Max 400 words.

Word limit NOT to be followed for diagram, numerical, derivation.

- Q.1 (a) (i) Computer uses which number system to perform calculations and store data: 02
 1. Hexadecimal 2. Decimal 3. Octal 4. Binary
 (ii) What type of programming is supported by Python:
 1. Object-Oriented-Programming 2. Structured Programming
 3. Functional Programming 4. All of the above
 (b) The correct extension of python file is _____. 02
 (c) List the standard data types in python with one example in each. 03
OR
 What is a variable in python? What are the rules for naming a variable? Explain by giving examples. 03
 (d) Write a Python program to check whether a number is prime or not prime. 05
OR
 What are the various arithmetic and logical operators that python support? Write syntax for each operator. 05
- Q.2 (a) (i) Which of the following operators has its associativity from right to left? 02
 1. + 2. // 3. % 4. **
 (ii) What will be the output of the following code:

```
n=7
c=0
while(n):
    if (n>5):
        c = c+n-1
        n = n-1
    else:
        break
print(n)
print(c)
```

 1. 5 11 2. 5 9 3. 7 11 4. 5 2

- (b) What will be the output of the following Python code?

```
for i in range(0,2,1):
    print("Hello")
```

 03
 (c) Explain the various control statements in Python with syntax. 03
OR
 Write a program to check whether a person is eligible to vote or not. Take age as input from user. 03
 (d) Write the output of the following code: 05

```
x=5
while (x<15):
    print(x**2)
    x+=3
```

OR
 Write a program to print the table of a number accepted from user. 05
- Q.3 (a) (i) What will be the output of:

```
str1 = "PYthon class"
print(str1[1:4],str1[0:-1])
```

 1. Yth PYthon clas
 2. PYt Python clas
 3. PYth Python clas
 (ii) What will be the output of:

```
str1 = "Python Lab"
print(len(str1))
```

 1. 9 2. 10 3. 11 4. 12 02
 (b) Write the output of the following:

```
str1="PYTHON"
Print(str1.lower())
```

 02
 (c) How to split strings and what function is used to perform that operation? 03
OR
 Write syntax to take a input from a user and typecast it to an integer. 03
 (d) What is a directory? Which module in python allows to use various methods with directory. Write the syntax for the following: (i) to get current directory in Python 05
 (ii) change directory in python (iii) making a new directory in python

- Q.4 (a) What is the output : 02
`colors=["red","blue","green","pink"]`
`Print(colors[2])`
- (b) What is the output : 02
`colors=["red","blue","green","pink"]`
`Print("yellow" in colors)`
- (c) What is the difference between tuples and list? 03
- OR**
- Explain how to create a dictionary in python. 03
- (d) Explain indexing and negative indexing in tuple with example. 05
- OR**
- Find the output of the following: 05
`L1 = [100,900,300,400,500]`
`START = 1`
`SUM = 0`
`for C in range(START,4):`
`SUM = SUM + L1[C]`
`print(C, ":", SUM)`
`SUM = SUM + L1[0]*10`
`print(SUM)`
- Q.5 (a) What is the output of the following code: 02
class Student:
`def __init__(self,name,id):`
`self.name=name`
`self.id=id`
`print(self.id)`
`std=Student("Simon",1)`
`std.id=2`
`print(std.id)`
- (b) In python, what is method inside class? 02
1. Attribute 2. Object 3. Argument 4. function

- (c) Explain operator overloading with an example in python. 03
- OR**
- What is the output of the following code: 03
`class A():`
`def disp(self):`
`print("A disp()")`
`class B(A):`
`pass`
`obj = B()`
`obj.disp()`
- (d) Explain exception handling with example. 05
- OR**
- Explain Method Resolution order in Python with an example. 05



ITC-101

Examination – May- 2024

B.Tech. I/II Sem: Common to all Branches
Python Programming

Time: 2 Hrs

Max. Marks: 60

Min. Marks: 19

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Word limit be observed as follows:

Part a & b – Objective type/Fill in the blanks/One sentence answer.

Part c – Max 100 words and Part d – Max 400 words.

Word limit NOT to be followed for diagram, numerical, derivation.

		Marks	BL	CO
Q.1	(a) Fill in the blank: In Python, the += operator is used for _____ assignment.	02	1,2	1
	(b) Define interactive shell and IDLE.	02	1,2	1
	(c) Explain the difference between mutable and immutable variables in Python. Provide examples to illustrate each.	03	1,2	1
	OR			
	Describe the numerical types available in Python and their respective use cases. Provide examples of each type.	03	1,2,3	1
	(d) Discuss the importance of indentation and comments in a Python program. How do they enhance code readability and maintainability? Provide examples to illustrate their usage.	05	1,2,3	1
	OR			
	Explain the concept of operators and expressions in Python. Provide examples of arithmetic, comparison, and logical operators along with their use in expressions.	05	1,2,3	1
Q.2	(a) The elif statement is short for _____ and is used to check for multiple conditions after an initial if statement.	02	1,2	2
	(b) Define a range in Python.	02	1,2	2
	(c) Explain the difference between the break, continue, and pass statements in Python. Provide examples to illustrate their usage.	03	1,2,3	2
	OR			
	With the help of an example describe the concept of nested loops in Python.	03	1,2,3	2
	(d) Write a Python program to find the sum of all numbers between 1 and 100 that are divisible by 3.	05	1,2,3,6	2
	OR			
	Write a Python program to find the factorial of a given number using a for loop.	05	1,2,3,6	2
Q.3	(a) Fill in the blank: In Python, the _____ operator is used to access individual characters in a string by their index positions.	02	1,2	3
	(b) Define slicing in Python strings.	02	1,2	3

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	(c) Explain the process of converting a string to a number and vice versa in Python. Provide example to illustrate this conversion.	03	1,2,3	3
	OR			
	Describe the role of the 'os' and 'sys' modules in Python for manipulating files and directories. Explain with example.	03	1,2,4	3
	(d) Write a Python program to read numbers from a text file, calculate their sum, and write the result to another file.	05	1,2,4,6	3
	OR			
	Discuss the significance of the CSV (Comma-Separated Values) and tab-separated file formats in Python for handling structured data. Provide examples to demonstrate their usage.	05	1,2,3	3
Q.4	(a) In Python, a list is a collection of _____ separated by commas and enclosed within square brackets.	02	1,2	4
	(b) Define a tuple in Python.	02	1,2	4
	(c) Explain the difference between a list and a tuple in Python. When would you use a list over a tuple, and vice versa?	03	1,2,4	4
	OR			
	Discuss the key characteristics of dictionaries in Python and their significance in storing and retrieving data. Provide examples to illustrate the usage of dictionaries.	03	1,2,3	4
	(d) Write a Python program to sort a list of numbers in ascending order using the 'sorted()' function.	05	1,2,3,6	4
	OR			
	Write a Python program to create a dictionary containing student names as keys and their corresponding scores as values. Then, calculate the average score of the students.	05	1,2,3,4,6	4
Q.5	(a) Fill in the blank: In object-oriented programming (OOP), a class is a blueprint for creating _____ in Python.	02	1,2	5
	(b) Define inheritance in the context of object-oriented programming.	02	1,2	5
	(c) Explain the concepts of overloading and overriding in Python classes. Provide example to illustrate each concept.	03	1,2,4	5
	OR			
	Explain the concept of data hiding in object-oriented programming. How does it contribute to encapsulation? Provide an example to illustrate.	03	1,2,3	5
	(d) Write a Python program to create a class called Rectangle with attributes length and width. Include methods to calculate the area and perimeter of the rectangle.	05	1,2,3,6	5
	OR			
	Discuss the importance of exception handling in Python programming. Explain the purpose of the try, except, and finally clauses in handling exceptions. Provide examples to demonstrate exception handling.	05	1,2,3	5

Bloom's Taxonomy Levels (BL): 1-Remembering, 2-Understanding, 3-Applying, 4-Analysing, 5-Evaluating, 6-Creating

Paper Code: ITC-101
Examination – May- 2025
B.Tech.II Sem : Information Technology
Subject Name: Python Programming

Time : 3 Hrs

Max. Marks : 60
Min. Marks : 19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c & d has internal Choice. Assume missing data, if any.

Word limit be observed as follows:

Part a(i) and a(ii) Objective type/Fill in the blanks/true& false.

Part (b) – Three/four sentences based answer question (Maximum 50 words)

Part (c) – Short Descriptive Ans./Numerical/ algorithm /programs based Question. (Maximum 100 words)

Part (d) – Long Answer /Shorts -notes /Descriptive Ans. based Question (Maximum 400 words)
Word limit NOT to be followed for diagram, numerical, derivation.

		Marks	BL	CO
Q.1 (a)-I	What is the default Python interpreter shell called?	01	1	1
	a) CMD b) Bash c) IDLE d) Notepad++			
(a)-II	Which of the following statements about variables in Python is true?	01	2	1
	a) Variables in Python need explicit type declarations. b) Variables must be declared before they are assigned. c) A variable can change its data type during execution. d) Python does not support variables.			
(b)	What is an immutable variable? Give a Python example.	02	2	1
(c)	What is the difference between the interactive shell and script mode?	03	3	1
	OR			
	Describe the different types of data representations in computers with examples.	03	3	1
(d)	Explain the role of variables in Python programming. How does Python handle variable type assignment internally?	05	5	1
	OR			
	Write a python program that demonstrates the use of all logical operators.	05	6	1
Q.2 (a)-I	What will be the result of this expression: True and False or True?	01	2	2
	a) True b) False c) None d) Error			
(a)-II	Which loop is guaranteed to execute at least once?	01	2	2
	a) for loop b) while loop c) do-while loop d) None			
(b)	Define Boolean logic and give an example in Python.	02	3	2
(c)	Explain what the range() function does in a for loop.	03	3	2
	OR			
	Write a short code to check if a number is positive, negative, or zero.	03	3	2
(d)	Discuss the purpose of break, continue, and pass statements with one example for each.	05	5	2
	OR			
	Write a program using if-elif-else to assign grades based on marks. 90-100: A, 80-89: B, 70-79: C, below 70: Fail	05	6	3
Q.3 (a)-I	Which module helps in interacting with the file system?	01	1	3

	a) math b) file c) os d) random			
(a)-II	What does open('data.txt', 'w') do?	01	1	3
	a) Opens file for reading b) Opens file for writing, creates if not exists c) Appends data to file d) Deletes the file			
(b)	Define indexing in python.	02	2	3
(c)	Write the syntax to convert a string "45.6" into a float.	03	4	3
	OR			
	Write a program to slice the string "PythonProgramming" from index 6 to 16.	03	4	3
(d)	How can strings be converted to numbers and vice versa? Provide code examples.	05	5	3
	OR			
	What is a CSV file? Explain how to write and read a CSV file using Python.	05	5	4
Q.4 (a)-I	Which method is used to add an element at the end of a list?	01	1	4
	a) add() b) append() c) insert() d) extend()			
(a)-II	Which of the following can be a key in a Python dictionary?	01	2	4
	a) list b) dictionary c) tuple d) set			
(b)	Write the syntax to insert an element at the second position in a list.	02	3	4
(c)	What is the difference between a list and a tuple in Python?	03	4	4
	OR			
	How is data stored in a dictionary different from a list?	03	4	4
(d)	Explain the differences between lists, tuples, and dictionaries with examples.	05	5	4
	OR			
	Describe various list operations such as append, insert, remove, pop, and slicing with examples.	05	5	4
Q.5 (a)-I	What is the correct way to define a class in Python?	01	1	5
	a) function MyClass(): b) class MyClass: c) define class MyClass: d) object MyClass:			
(a)-II	What does method overloading mean in Python?	01	2	5
	a) Redefining the same method name in subclass b) Defining multiple methods with the same name but different parameters c) Defining multiple constructors d) None of the above			
(b)	What is a class in Python?	02	2	5
(c)	Define exception handling in Python.	03	3	5
	OR			
	What is method overriding? Give an example.	03	3	5
(d)	Explain object-oriented programming concepts with an example.	05	5	5
	OR			
	What is inheritance? How do you implement inheritance in Python?	05	5	5